Jaypee Institute Of Information Technology
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Data Structures Lab (15B17CI371)

Mini Project Synopsis

Project title:

<u>SoundWave:</u>
<u>A Dynamic Music Streaming Platform</u>

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Description of the project

Overview :

SoundWave is a console-based music streaming platform designed to provide a personalized listening experience with basic features such as user authentication, playlist management, offline listening, and music recommendations. The system uses core Data Structures and Algorithms (DSA) concepts to store and manage user data, music tracks, and playlists efficiently. The platform allows users to search and sort songs, create custom playlists, and receive basic recommendations based on their listening habits.

Workflow :

User Authentication

- Users can sign up with a username and password. Passwords are securely hashed and stored.
- Users can log in to the system, and their credentials are validated using file handling.

Data Storage & Data Structures

- **Music Library**: Songs are stored in an array or linked list, with attributes such as title, artist, and genre.
- **Playlists**: User-created playlists are stored in linked lists, enabling dynamic modifications (add/remove songs).
- **Recommendation System**: Songs are recommended based on user history using graph-based traversal and heap algorithms for priority ranking.

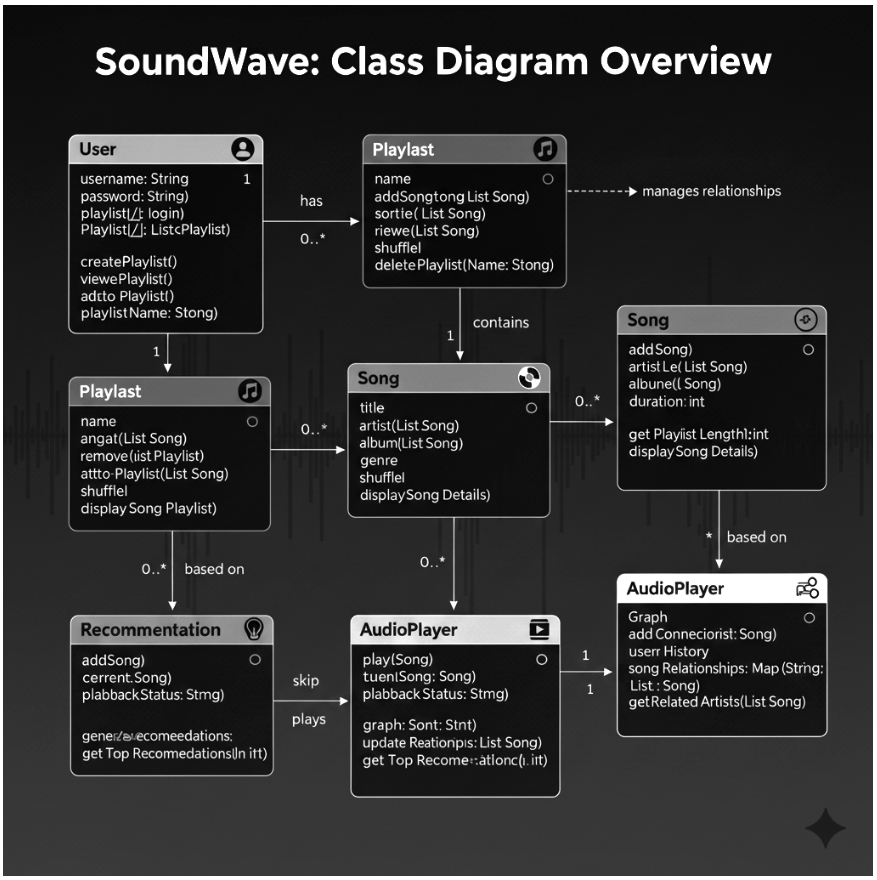
Operations (Functionalities)

- **Create Playlist**: Users can create and manage personalized playlists.
- **Search**: Users can search songs by title, artist, or genre using linear or binary search algorithms.
- **Sort**: Users can sort their playlists by title, artist, or release year using sorting algorithms like Merge Sort or Quick Sort.
- **Recommendations**: Based on user history, the system suggests songs from similar artists or genres.
- **Offline Listening**: Users can download songs for offline use, with data stored in text files for persistence across sessions.

Use of Data Structures :

- **Arrays/Linked Lists:** To store songs and user playlists dynamically.
- **Heap/Merge Sort:** Used to sort playlists by different criteria such as release date or popularity.
- **Graphs:** For recommending songs based on relationships between artists, albums, or genres.
- **File Handling:** For persistent storage of user data, playlists, and music catalog.
- **Hash Tables:** Used for fast searching of song information.

Class Diagram :



Real-Life Applications of the Project :

In today's digital music landscape, managing large music libraries and personalizing music recommendations is a significant challenge. Traditional music players often fail to provide personalized listening experiences. By implementing this **SoundWave** music streaming platform using efficient data structures (arrays, linked lists, graphs, heaps), this system offers:

- **Faster access** to a vast collection of songs through optimized search and sorting.
- **Personalized playlists** and music recommendations based on user interactions and song relationships.
- **Offline listening** for users, ensuring that their playlists and music preferences are always accessible, even without internet connectivity.

SoundWave provides an innovative way to handle large-scale music data efficiently, solving common problems faced by traditional music streaming platforms.